MVP + Suggested Prize Tracks

The idea in one line

A Chrome extension that lets you learn a language from the media you already watch. On YouTube it overlays **color-aligned bilingual captions** ("how" matches "cómo", "are you" matches "estás"), explains words in a sidebar, speaks them aloud with ElevenLabs, and tracks your progress in a Tiger Data dashboard.

The problem

Immersion is the best way to learn a language, but:

- Living abroad isn't possible for most people.
- Foreign media is hard to access legally (VPNs, region locks).
- Switching to media you don't normally watch lowers engagement. Not everyone wants to watch soap operas to learn Spanish.

Our answer: keep watching what you love. The extension turns it into a language lesson.

Unifying story

> **Speak it, track it, improve it.** Every pronunciation attempt becomes a data point, and the dashboard shows your accent improving over time.

Watch, understand, practice, track progress.

MVP definition

A Chrome extension that, on any YouTube video, overlays color-aligned captions in your target language and lets you click a word for an explanation and audio pronunciation. You can shadow a line into the mic, get a score, and see it land live in a dashboard.

Must have (the demo fails without these)

- [ ] Grab the video's captions (YouTube only)
- [ ] Translate them with Gemini and align words/phrases between the two languages
- [ ] Render a custom caption overlay with matching colors across both lines
- [ ] Click a word or phrase to open a sidebar with a gloss, a grammar note, and an example
- [ ] ElevenLabs pronunciation (text-to-speech) for words and full lines
- [ ] Shadowing: record yourself, transcribe with ElevenLabs Scribe, score and highlight missed words
- [ ] Events logged to Tiger Data, plus a dashboard with a live-updating chart

Should have

- Save-to-vocab list
- Continuous aggregates, compression, heatmap, and trouble-words panels
- Level-adaptive captions (known words stay plain)
- Tone and culture notes (tú vs. usted, slang, idioms)

Cut for now

- Netflix (DRM and custom subtitle delivery could eat the whole weekend)
- Accounts, auth, and a polished settings UI
- More than 1-2 target languages. Start with Spanish. The pipeline should stay language-agnostic by default, since Gemini handles other languages.

Stretch ideas (only after the core works)

- **Reverse mode:** video in the target language, overlay glosses into English. Same pipeline with source and target flipped.
- **Select-any-text mode:** run the aligned translation on any webpage via a floating tooltip. This makes the "works anywhere" claim true.
- **Voice tutor:** an ElevenLabs conversational agent that chats with you in Spanish about the video.
- **"What's that?" visual mode:** click the video, send the frame to Gemini, and get the object named in the target language.
- **AI dubbing:** pre-generate target-language audio with ElevenLabs and duck the original. Fiddly to sync.
- **Joke "brainrot translator" mode** for the goofy-project track.

Suggested prize tracks

Primary targets (we are going for these)

Track	Prize	Why we fit	What to show judges
Best Use of ElevenLabs	Wireless earbuds	TTS, speech-to-text scoring, and a slow mode form an interactive speaking loop, not just a play button.	Hear a word pronounced, shadow a line, see the score. Name every ElevenLabs product used in the writeup.
Best Use of Tiger Data	Stream Deck Mini	Every lookup and shadowing attempt becomes time-series data. We use hypertables, continuous aggregates, and compression.	Live dashboard tick on a fresh attempt, plus trouble words and a heatmap served from aggregates.

Comes naturally

Track	Prize	Notes
Best Use of Gemini API	MLH swag kits	Gemini is the core engine: alignment, grammar notes, quiz and summary generation.
Best AI Hack	Apple Watch SE3	Requirement is that the project features a model. We use LLMs and speech models meaningfully.
Best Overall Hack (1st/2nd/3rd)	Mac Mini + Full Sail full-ride / Samsung Odyssey / keyboard + mic	Only if the demo is smooth and the pitch lands. Our pitch is strong: immersion without changing your media.

Principles

- **A working colored overlay beats a half-working everything.** If the core milestone slips, cut the Should-haves.
- **Every feature must be demoable in under 15 seconds, or cut it.** Judges see a 90-120 second demo, not a feature list.
- **No forced integrations.** Each sponsor tool must earn its place in the product.

Team roles

Person	Focus
Extension lead (you)	Core overlay, sidebar, shadowing UI (record button and diff display). The extension is the demo centerpiece.
Person B (audio + AI)	Alignment prompt quality, ElevenLabs TTS and STT, scoring logic, Gemini notes. Hand finished endpoints to the extension lead early.
Person C (data)	Backend, Tiger Data schema, aggregates, seed script, dashboard.

All three work together on the core overlay milestone (hours 2-12).

Risks

- **Alignment quality** is make-or-break. Test the prompt on slang, idioms, and fast speech early.
- **Latency:** solved by pre-translating and caching, but test on a long video.
- **YouTube DOM changes and auto-generated captions:** auto-captions have no punctuation and are messy. Demo on a video with clean manual captions.
- **Microphone access from a content script** may force a design change. Test early.
- **Scope creep:** every new platform or language is a rabbit hole.